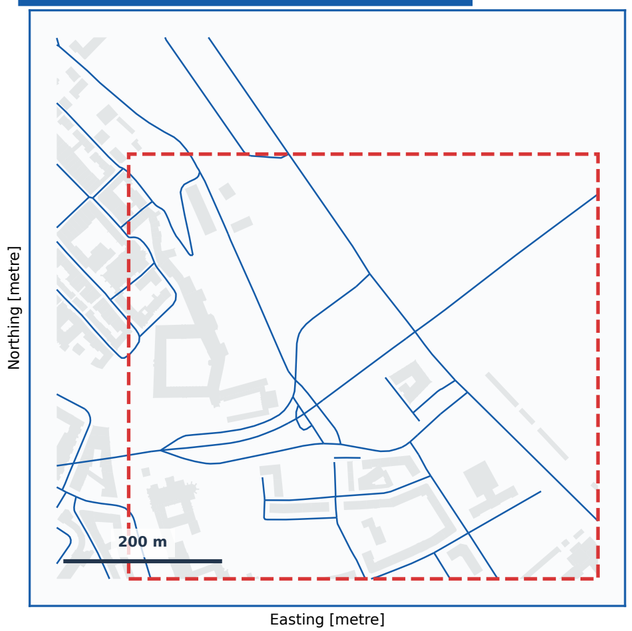
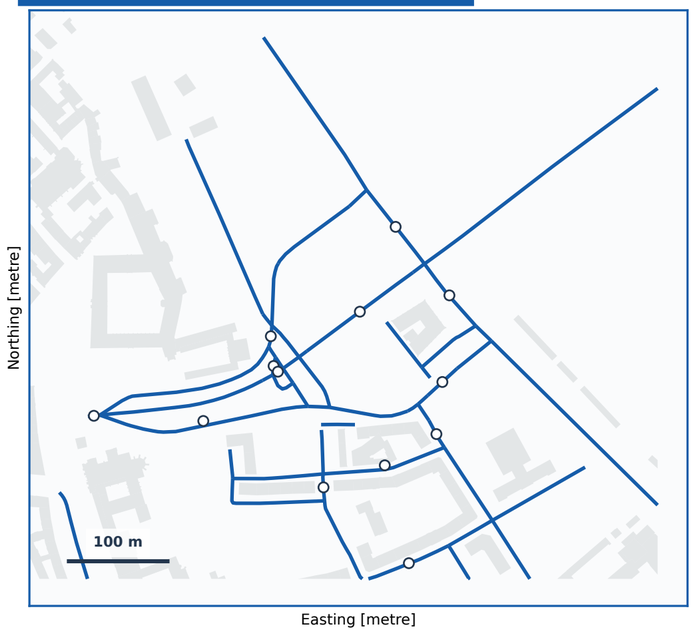
Road-Segment Graph Construction and Spatial Neighborhood Encoding

Real Warsaw roads | k=6 nearest segment centroids | maximum radius=250 m

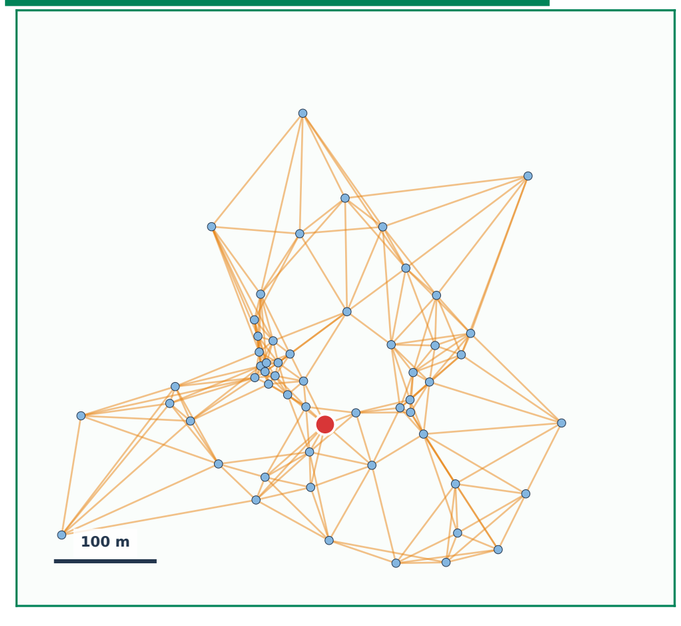
(a) Detailed source road network



(b) Road segments and centroids



(c) kNN proximity graph (k=6, r=250 m)



(d) Edge construction procedure

1. Segment centroids

Compute centroid c_i for each source road.

2. Neighbor search

Find k=6 nearest centroids within 250 m.

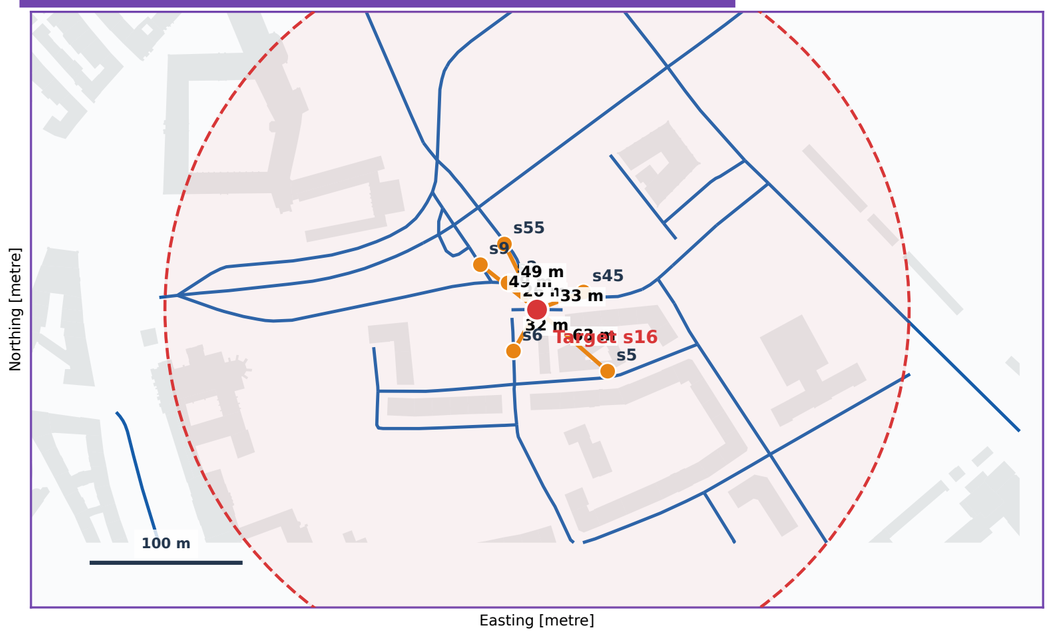
3. Undirected graph

Connect i and j if either selects the other.

4. Node feature matrix

Attach road and graph-context attributes.

(e) Real spatial neighborhood and centroid distances



Real road segment | Target centroid | Neighbor centroid | Spatial kNN edge | 250 m radius

All maps use actual source geometries. Edges represent centroid proximity, not verified road-junction connectivity.